

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 4372	Course Title: Verilog HDL and Programmable Logic Devices
Semester: 4	Credits: 4
Course Category: Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide awareness on the basics of Hardware Description Language,
- Programmable Logic Devices and its use in designing Digital Electronics Circuits.

Course Prerequisites:

Topic	Course Code	CourseTitle	Semester
Electronics Components and Circuits	3043	Electronic Circuits	3
Combinational and Sequential digital circuits	3044	Digital Electronics	3
Basics concepts of programming	1008	Introduction to IT systems Lab	1
		Python programming lab	3

Course Outcomes:

On completion of the course, the student will be able to,

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basics of Hardware Description Language and program structure	14	Understanding
CO2	Make use of gate level modeling and data flow modeling to design digital circuits	15	Applying
CO3	Make use of Behavioral Modelling to design combinational and sequential circuits	15	Applying
CO4	Outline the architecture of CPLDs and FPGA	14	Understanding
	SeriesTest	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3		2				
CO3	3		2				
CO4	2						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basics of Hardware Description Language and program structure		
M1.01	Explain the evolution of Computer Aided Design, HDL, and importance of HDL	3	Understanding
M1.02	Explain hierarchical modeling concepts	4	Understanding
M1.03	Explain Design block–stimulus-data types	4	Understanding
M1.04	Explain the concept of Verilog Modules and Ports	3	Understanding
Contents: Evolution of Computer Aided design, Overview of Digital design with Verilog HDL - Evolution of CAD - emergence of HDLs, typical HDL based design flow- why Verilog, trends in HDLs. Hierarchical Modelling Concepts- Top- down and bottom-up design methodology, differences between modules and module instances, components of a simulation. Design block, stimulus block, basic concepts, Lexical conventions, data types, system tasks, compiler directives. Module-Components of a Verilog module, Ports- definition, list of ports, port declaration, port connection rules			
CO2	Make use of gate level modelling and dataflow modelling to design digital circuits		
M2.01	Identify logic gates as predefined primitives	1	Understanding
M2.02	Illustrate the instantiation of basic gates	1	Understanding
M2.03	Make use of gate level modelling to design different Combinational logic circuits	3	Applying
M2.04	Describe gate delays, minimum value, typical value and maximum value.	1	Understanding

M2.05	Describe continuous assignment statement and implicit continuous assignment statement	1	Understanding
M2.06	Explain regular assignment delay, implicit continuous assignment delay and net declaration delay	1	Understanding
M2.07	Describe the basis of data flow modelling, expressions operators and operands	3	Understanding
M2.08	Make use of dataflow modelling to design combinational and sequential logic circuits	4	Applying
	SeriesTest1	1	
Contents: Gate level modelling - Instantiation of basic gates, design of simple combinational circuits using gate level modelling life multiplexer, full adder and ripple carry adder, gate delays, gated delays, minimum value, typical value and maximum value Data flow modelling - continuous assignment statement, implicit continuous assignment statement, regular assignment delay, implicit continuous assignment delay and net declaration delay, expressions operators and operands, different types of operators, design of simple combinational and sequential circuits - full adder, multiplexer, d flip flop, ripple counter			
CO3	Make use of Behavioral Modelling to design combinational and sequential circuits		
M3.01	Explain the significance of structured procedures always and initial in behavioral modelling	3	Understanding
M3.02	Explain procedural assignments.	3	Understanding
M3.03	Identify conditional statements used in behavioral modelling	3	Applying
M3.04	Identify the looping statements used in Verilog,	3	Applying
M3.05	Develop Verilog programs to realize simple combinational and sequential circuits using behavioral modelling	3	Applying
Contents: Behavioral Modelling - Structured procedures, initial and always, blocking and non-blocking statements - explanation with examples. Delay control, event control, conditional statements, looping statements, simple design examples (4-1 Multiplexer, 4-bit counter etc..)			
CO4	Outline the architecture of CPLDs and FPGA		
M4.01	Explain the architecture of PAL and PLA	4	Understanding
M4.02	Explain CPLD with the help of block schematic	3	Understanding

M4.03	Illustrate the architecture of FPGA	4	Understanding
M4.04	Explain configurable logic blocks (CLBs)	1	Understanding
M4.05	Compare FPGA with CPLD	2	Understanding
Test2		1	
Contents: Simple Programmable Logic Devices, Architecture of PAL and PAL ,compare PAL and PLA, Complex Programmable Logic Devices-Architecture and applications, Field Programmable Gatearrays-Architecture and applications, comparison of CPLD and FPGA, Concept of configurable logic blocks			

Text/Reference Books:

T/R	Book Title/Author
T1	VerilogHDL–SamirPalnitkar-PearsonEducation
T2	DigitaldesignMorrismano,ThirdEdition,PHI
R1	Digital Design With an Introduction to the Verilog HDL, VHDL, andSystemVerilog SixthEdition ByPearsonPaperback–2018byM.MorrisMano, <u>Michael D.Ciletti</u>
R2	FPGAbasedsystemdesignWayneWolf,Pearson Education
R3	T.R.Padmanabhan,B BalaTripura Sundari, Design Through Verilog HDL,Wiley2009.
R4	FPGA prototyping by Verilog examples–PONGP Chu-Willey
R5	HDL Programming- Nazeih M Botros- Dream Tech
R6	Digital Systems Design with FPGA sand CPLDs- Ian Grout, Elsevier

Online Resources:

Sl.No	WebsiteLink
1	https://nptel.ac.in/courses/106/105/106105165/
2	https://www.tutorialspoint.com/vlsi_design/vlsi_design_verilog_introduction.htm
3	http://www.ee.ic.ac.uk/pcheung/teaching/ee2_digital/fpga%20&%20cpld%20tutorial.pdf
4	https://www.youtube.com/watch?v=PJGvZSlLKs
5	https://www.youtube.com/watch?v=pR6-aNxHNac